Exp: 9 **Subnetting in Cisco Packet Tracer**

Aim:

To design multiple subnets, assign static IP addresses, connect the subnets through a router and simulate packet transmission with and without the router.

Requirements:

* Cisco Packet Tracer
* 2 switches, 1 router (2811/1941), 6 PCs

Addressing Plan (example: 192.168.10.0/24 split into 2 subn

* Subnet 1: 192.168.10.0/25 - hosts 192.168.10.1 to .126, gateway 192.168.10.1 (router)
* Subnet 2: 192.168.10.128/25 - hosts 192.168.10.129 to .254, gateway 192.168.10.129 (router)
* Subnet mask 255.255.255.128

Procedure:

1. Place 3 PCs on each switch and connect each switch to the router (switch to ~~router~~ router interface Fa0/0 and Fa0/1).
2. Assign static IPs and the default gateway to each PC, e.g. PC1 192.168.10.2 (gw 192.168.10.1), PC4 192.168.10.130 (gw 192.168.10.129).
3. Configure the router interfaces:

```
Router> enable
Router# configure terminal
Router(config)# interface fastethernet0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.128
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface fastethernet0/1
Router(config-if)# ip address 192.168.10.129 255.255.255.128
Router(config-if)# no shutdown
```

```
Router(config-if)# end
Router# show ip route
```

4. (a) With the router connected: ping from a PC in subnet 1 to a PC in subnet 2 - replies are received.
5. (b) Remove / disconnect the router and ping again - the request fails (Destination host unreachable / request timed out) because different subnets cannot communicate without a router.
6. Observe the PDU path in Simulation mode.

Result:

Subnets were designed and configured; communication across subnets succeeded only when connected through a router.